CSE-4621 Machine Learning (Set 001101) **Student id:**
Quiz # 1 **Time: 25 min**

1. Compare SGD, batch gradient descent, and mini-batch gradient descent in terms of Convergence speed, Computational efficiency, Noise in updates. You are only allowed to write the following words in the table: ***Slow, Fast, Low, High and Moderate***. --3

Method	Convergence Speed	Computational Efficiency	Noise in Updates
Batch GD	Slow	Low	Low
Mini-batch GD	Moderate	Moderate	Moderate
SGD	Fast	High	High

2. A student engineers a new feature for linear regression: $x3 = x1 \times x2$ and then applies standardization (Z-score: $\frac{x-\mu}{\sigma}$) to all features, including $x3$. --4
   a. Why is this problematic?
   b. Propose a better feature engineering and/or scaling strategy for interaction terms.

The interaction term x1×x2 **loses its meaning** after independent standardization.

Standardize the original features x1 and x2 first, then construct the interaction term x3.

3.

Consider a simple linear regression model:

$$\hat{y} = wx + b$$

Given a dataset with m examples $\{(x^{(i)}, y^{(i)})\}_{i=1}^{m}$, the **squared error loss** is defined as:

$$J(w,b) = \frac{1}{2m}\sum_{i=1}^{m}\left(\hat{y}^{(i)} - y^{(i)}\right)^2 = \frac{1}{2m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)^2$$

--8

a. Derive the partial derivative of the cost function J(w,b) with respect to w and b.

For w:	For b:
$\frac{\partial J}{\partial w} = \frac{1}{m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)x^{(i)}$	$\frac{\partial J}{\partial b} = \frac{1}{m}\sum_{i=1}^{m}\left(wx^{(i)} + b - y^{(i)}\right)$

b. Write the gradient descent update rules for w and b using a learning rate α (alpha).

For w:	For b:
$w := w - \alpha\frac{\partial J}{\partial w},$	$b := b - \alpha\frac{\partial J}{\partial b}$

c. Given the following data: input X = [1, 2, 3], output Y= [2, 3, 5], and initial parameters w=0, b=0, perform one step of gradient descent with α=0.1.

$x^{(i)}$	$y^{(i)}$	$\hat{y}^{(i)} = wx^{(i)} + b$	Error $\hat{y}^{(i)} - y^{(i)}$
1	2	$0 \cdot 1 + 0 = 0$	-2
2	3	$0 \cdot 2 + 0 = 0$	-3
3	5	$0 \cdot 3 + 0 = 0$	-5

Compute gradients:

$$\frac{\partial J}{\partial w} = \frac{1}{3}[(-2)\cdot 1 + (-3)\cdot 2 + (-5)\cdot 3] = \frac{1}{3}[-2 - 6 - 15] = \frac{-23}{3} \approx -7.67$$

$$\frac{\partial J}{\partial b} = \frac{1}{3}[-2 - 3 - 5] = \frac{-10}{3} \approx -3.33$$

Apply updates (with $\alpha = 0.1$):

$$w := 0 - 0.1 \cdot (-7.67) = 0.767, \quad b := 0 - 0.1 \cdot (-3.33) = 0.333$$